

DATA ANALYSIS REPORT

BICT242: Data Scalability and Analytics

Student Course Recommendation System

1. Introduction

This project develops a recommendation system for an educational platform. The purpose is to recommend courses to students according to their interests and learning level. The project follows the BICT242 requirement to collect data, preprocess it, perform feature engineering, select a recommendation approach, evaluate the model and deploy it as an application.

2. Problem Statement

Students can have difficulty deciding which courses or learning areas are relevant to their interests. A recommendation system can reduce this difficulty by analyzing information about courses and a student's stated interests and then returning courses with similar content.

3. Project Objectives

- Create a dataset containing course information and simulated student profiles.
- Clean and prepare the data for recommendation.
- Convert textual course information into numerical features.
- Build a content-based recommendation model using TF-IDF and cosine similarity.
- Evaluate the recommendations using Precision@3, Recall@3 and F1@3.
- Deploy the model through a simple Streamlit web application.

4. Data Collection

For this prototype, a synthetic dataset was created because the course brief does not provide a required real-world dataset. The course dataset contains 20 courses. Each course has an ID, name, category, description and difficulty level. A second dataset contains 20 simulated student profiles with interests, levels and a target course used for evaluation.

Course categories include Programming, Web, Database, Data Science, AI, IoT, Networking, Security, Mobile, Cloud, Software and Design.

5. Data Description

Field	Meaning	Example	Type	Use
course_id	Unique course identifier	C01	Text	Identification
course_name	Name of the course	Python Programming	Text	Recommendation
category	Main subject area	Programming	Text	Feature
description	Short description of course content	Python programming...	Text	Feature
level	Difficulty level	Beginner	Categorical	Feature

6. Data Preprocessing

The CSV files were checked to make sure the required columns were present.

Text fields were combined into one course text fields.

Course names, categories, descriptions and levels were used as model input.

TF-IDF automatically represents important words numerically while reducing the influence of very common words.

Student interests are transformed using the same TF-IDF vocabulary as the course data.

7. Feature Engineering

The main engineered feature is a combined text representation made from course name, category, description and level. This allows the system to compare a student's interest profile with the content of each course. A small level-match bonus is also applied when the student's selected level matches the course level.

8. Data Analysis Summary

The dataset contains 20 courses and 20 simulated student profiles. The course catalogue covers 13 broad categories and three difficulty levels. This variety gives the prototype enough different course descriptions for a meaningful demonstration of text-based recommendation.

9. Limitations of the Data

The dataset is synthetic and therefore does not represent actual student behaviour. The student profiles use stated interests rather than long-term click, enrolment or rating histories. The dataset is relatively small compared with a production recommendation platform. The system currently recommends courses using content similarity and does not learn from collaborative user behaviour.

10. Conclusion

The data preparation stage produced a clean and structured dataset suitable for a content-based recommendation system. The main useful information was about the textual relationship between student interests and course descriptions. The prepared data was then used in the model selection and evaluation stage.